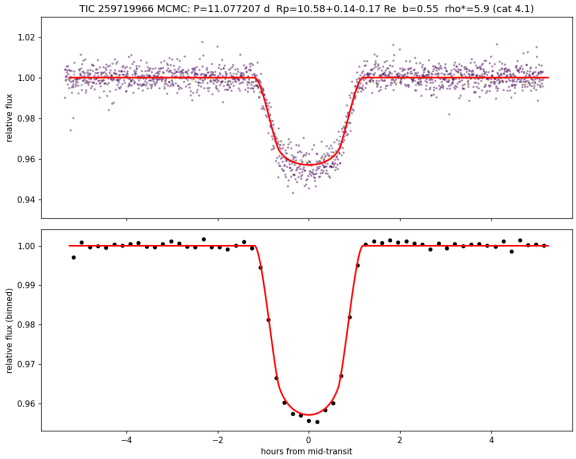
# TIC 259719966: observer brief

Transiting Jupiter-radius companion candidate around a mid-M dwarf. Prepared 2026-10-25 by Kishore Hariharan (Generative Experiences / tess-hunt) from the multi-sector fit in report.md. Research Note under review at RNAAS (AAS82771); not a TOI or CTOI as of 2026-10-25. Code and full vetting: https://github.com/khariha/tess-hunt




Phase-folded TESS photometry, all nine sectors, with the posterior-median transit model.

## Target

| | |
|---|---|
| TIC | 259719966 |
| Gaia DR3 | 6403128632917009792 |
| RA, Dec (J2000) | 322.67290, -63.61645 (21h30m41.5s, -63d36m59s) |
| Constellation | Indus |
| T / G | 13.80 / 14.98 |
| Host (TIC v8) | Teff 3443 K, R = 0.495 Rsun, M = 0.493 Msun, d = 147.1 pc |
| Gaia DR3 | RUWE 1.08, non_single_star = 0, single source within 3 arcsec |

## Ephemeris (BJD_TDB)

| | |
|---|---|
| P | 11.077207 +/- 0.000001 d |
| T0 | 2460877.82312 +/- 0.00020 |
| Duration (T14) | 2.37 +/- 0.03 h |
| Depth (undiluted sectors) | 38,300 +1,000 -1,200 ppm |
| Rp/Rs | 0.196 +/- 0.003 |
| a/Rs | 37.8 +/- 0.9 |
| b | 0.55 +/- 0.04 |
| Rp | 10.58 Rearth = 0.942 Rjup (+/- 0.015) |
| a | 0.087 AU |
| T_eq (A = 0) | 396 K |
| Transit-derived rho_star | 5.9 +/- 0.4 solar (TIC: 4.1) |

Prediction uncertainty stays under 1 minute through 2027 (P error times cycle count).

Upcoming mid-transits, UTC (ingress and egress at -71 and +71 min):

| Epoch | Mid-transit |
|---|---|
| 42 | 2026-10-29 13:33 |
| 43 | 2026-11-09 15:24 |
| 44 | 2026-11-20 17:15 |
| 45 | 2026-12-01 19:06 |
| 46 | 2026-12-12 20:58 |
| 47 | 2026-12-23 22:49 |
| 48 | 2027-01-04 00:40 |
| 49 | 2027-01-15 02:31 |
| 50 | 2027-01-26 04:22 |
| 51 | 2027-02-06 06:14 |
| 52 | 2027-02-17 08:05 |
| 53 | 2027-02-28 09:56 |
| 54 | 2027-03-11 11:47 |
| 55 | 2027-03-22 13:38 |

Dec -64: circumpolar from Chile, South Africa and Australia; never rises north of +26 latitude.

## Evidence summary

- Detected in Sector 1 (30-min FFIs) in an independent BLS search of 38,620 Tmag 12-15 M dwarfs; recovered independently in all 9 sectors TESS has observed the star (1, 27, 67, 94, 95, 101-104), 20 transits, 2018-2026.
- Noise-normalized difference images in the 200-s sectors 94 and 104 peak on the target at 33 and 37 sigma. In the 30-min Sector 1 data the centroid shows a 6.4 sigma shift and the difference-image peak lies 0.7 px from the target at 10 sigma, consistent with the 21-arcsec pixels and the neighbour below rather than a displaced source; the 200-s sectors resolve it.
- Odd and even depths agree (-230 +/- 370 ppm). Secondary at phase 0.5: 20 +/- 540 ppm, which disfavours a companion hot enough to be seen in the TESS band.
- Duration consistent with the TIC stellar density within the b / a/Rs degeneracy.

## Neighbour

Eight Gaia DR3 sources within 2 TESS pixels. The brightest, 2.1 px (about 44 arcsec) from the target, carries 27% of the target's G flux and enters the automatic TESS aperture in three sectors, where the depth measures shallower. The sense of that effect (shallower when the neighbour is included) independently places the dip on the target. Ground-based apertures should exclude it; at 44 arcsec that is routine.

## What would settle the nature of the companion

Radius alone spans giant planet, brown dwarf and late-M star. Expected RV semi-amplitude for a circular, edge-on orbit around 0.49 Msun at 11.08 d:

| Companion mass | K |
|---|---|
| 1 Mjup | ~150 m/s |
| 13 Mjup | ~1.9 km/s |
| 80 Mjup | ~12 km/s |

A single reconnaissance spectrum excludes the stellar case; two epochs near quadrature (phases 0.25 and 0.75) with a red or near-infrared spectrograph on a 4 m to 8 m telescope (ESPRESSO, NIRPS, HARPS, MAROON-X) separate planet from brown dwarf. Host is faint for RV but not prohibitive: T = 13.8, and the expected signal is large.

A ground-based transit on the target at the predicted time, with the 44-arcsec neighbour excluded, would close the remaining blend question. The 3.8% depth is reachable with a 0.4 m telescope.

## Caveats

Stellar parameters are TIC v8; the companion radius inherits their error. No spectroscopic characterization of the host exists. Candidate status until a mass is measured.

## Figures (this directory)

s1_tic259719966_mcmc.png (phase fold and model), s1_tic259719966.png (per-sector folds), s1_tic259719966_pixels.png (difference images), s1_tic259719966_corner.png (posterior), s1_tic259719966_sector1_vetting.png.